

ULYANOVSK VOSTOCHNY, Russian Federation

WMO: 279621

Lat: 54.401N

Lon: 48.803E

Elev: 252

StdP: 14.56

Time Zone: 4.00 (E04)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-13.1	-25.3	1.4	-19.6	-18.6	2.0	-12.6	23.4	19.7	21.5	19.4	2.1	350	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	91.8	71.4	88.0	70.2	84.4	68.1	73.6	86.8	71.8	83.7	69.9	82.0	9.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	110.5	79.9	67.9	103.5	77.7	66.1	97.0	74.9	37.4	86.4	35.6	83.5	34.2	81.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.5	17.5	-26.5	97.3	6.3	3.5	-31.0	99.9	-34.7	101.9	-38.3	103.9	-42.8	106.4		
			-26.6	76.0	5.9	2.2	-30.9	77.6	-34.4	78.9	-37.7	80.1	-42.0	81.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	11.3	12.2	23.5	43.0	60.3	66.7	70.3	67.4	56.4	42.5	31.1	19.5	(6)
(7)		DBStd	23.03	12.83	12.49	9.91	8.90	7.62	6.93	5.90	7.14	6.84	7.96	8.25	11.10	(7)
(8)		HDD50	5123	1201	1057	822	250	8	1	0	0	19	254	566	945	(8)
(9)		HDD65	8803	1666	1477	1287	659	187	61	18	57	269	696	1016	1410	(9)
(10)		CDD50	2271	0	0	0	40	327	502	628	539	211	23	1	0	(10)
(11)		CDD65	477	0	0	0	0	41	112	181	132	11	0	0	0	(11)
(12)		CDH74	5470	0	0	0	24	625	1280	1946	1454	141	0	0	0	(12)
(13)		CDH80	2041	0	0	0	2	172	477	767	595	28	0	0	0	(13)
(14)	Wind	WSAvg	8.2	8.5	8.0	8.6	8.8	8.1	7.9	7.1	7.3	7.3	9.1	8.9	9.2	(14)
(15)	Precipitation	PrecAvg	19.6	1.4	1.1	1.0	1.3	1.6	2.4	2.0	2.1	1.7	1.5	1.3		(15)
(16)		PrecMax	26.8	2.9	2.1	2.5	3.5	3.7	4.8	5.2	4.5	5.6	3.0	3.5	3.4	(16)
(17)		PrecMin	13.4	0.5	0.4	0.0	0.1	0.1	0.1	0.1	0.5	0.4	0.7	0.4	0.5	(17)
(18)		PrecStd	3.1	0.6	0.5	0.6	0.8	0.9	1.3	1.3	1.1	1.5	0.7	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	37.4	48.2	77.2	88.1	95.0	96.9	97.1	84.1	66.4	52.2	39.4	(19)
MCWB			34.8	34.5	42.3	59.5	68.2	73.5	72.2	70.1	64.7	54.1	48.6	38.3	(20)	
2%		DB	34.0	35.3	41.2	71.4	84.4	89.8	91.6	91.6	77.5	62.4	48.2	37.2	(21)	
		MCWB	33.3	33.5	36.7	56.1	65.9	70.8	72.1	70.0	62.4	52.7	46.6	35.9	(22)	
5%		DB	32.0	33.4	37.8	66.0	81.0	85.7	87.8	86.3	73.5	57.5	44.9	35.4	(23)	
		MCWB	31.4	31.9	34.9	53.2	64.5	69.0	70.6	69.5	60.5	51.4	43.2	34.3	(24)	
10%		DB	28.6	29.8	35.9	59.4	77.0	81.0	84.1	82.3	69.7	53.9	42.5	33.6	(25)	
		MCWB	27.8	28.5	33.9	49.4	62.2	67.2	69.1	67.5	58.9	49.2	41.3	32.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	34.8	42.5	61.6	70.1	75.5	77.0	73.8	67.3	56.9	50.2	38.8	(27)
MDCB			35.3	36.0	47.5	75.8	82.4	91.0	91.3	89.4	79.1	61.8	51.8	39.1	(28)	
2%		WB	33.1	33.1	37.5	57.6	67.7	72.2	74.3	72.0	64.4	54.5	46.5	35.5	(29)	
		MDCB	33.5	34.3	39.7	68.6	81.9	85.9	86.4	85.9	73.5	60.3	47.8	36.1	(30)	
5%		WB	31.1	31.2	35.6	53.7	65.4	70.1	72.6	70.2	62.2	52.3	43.2	33.9	(31)	
		MDCB	31.7	32.6	37.4	64.3	78.4	82.3	83.6	83.6	70.8	57.0	44.3	34.4	(32)	
10%		WB	27.7	27.8	34.1	50.4	63.5	68.5	70.6	68.7	60.1	50.2	40.8	32.4	(33)	
		MDCB	28.4	29.2	35.8	58.8	75.5	79.7	81.2	80.5	67.1	53.2	41.7	33.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.9	15.7	16.6	19.2	22.7	20.3	21.2	20.8	18.1	12.3	8.4	10.3	(35)
MCDBR			9.6	12.3	15.2	29.3	29.9	28.3	28.1	28.7	26.2	19.6	10.9	7.3	(36)	
MCWBR			9.8	11.0	12.0	16.8	14.3	12.9	12.4	11.9	13.4	12.4	9.6	7.3	(37)	
5% WB		MCDBR	9.5	11.1	13.6	28.0	27.4	24.5	24.9	25.0	23.0	16.9	9.8	7.5	(38)	
		MCWBR	9.7	10.3	11.1	17.2	13.9	11.9	12.1	11.3	12.7	12.0	9.1	7.7	(39)	
																(39)
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.325	0.368	0.423	0.399	0.390	0.406	0.433	0.381	0.355	0.320	0.293	(40)	
(41)		taud	2.071	2.040	2.039	2.119	2.311	2.386	2.340	2.244	2.363	2.333	2.210	2.065	(41)	
(42)		Ebn at Noon	195	231	249	253	269	272	265	248	248	225	187	166	(42)	
(43)		Edh at Noon	22	32	41	43	38	36	36	38	30	24	19	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	252	569	1000	1348	1768	1896	1875	1496	947	464	196	142	(44)	
(45)		RadStd	38	70	122	118	132	178	159	129	128	81	33	36	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+203	N/A

Nomenclature: See separate page